EXP 1.5	A PYTHON PROGRAM TO SOLVE A LPP BY - GRAPHICAL METHOD

AIM:

To Solve a Linear programming problem by graphical method using Python.

PROCEDURE:

1. Install Required Libraries:

o Make sure that the librariesnumpy,scipy and matplotlib are installed. They can be

installed using the following command.

!pip install numpyscipy, matplotlib

2. Define the Problem:

o Identify the objective function to be maximized or minimized.

o Determine the constraints (equality and inequality).

3. Set Up the Problem in Python:

o Use the scipy.optimize.linprog function to set up and solve the LPP.

4. Run the Code:

o Execute the Python code to get the optimum solution.

5. Analyze the Output:

o Interpret the results and validate the solution against the problem constraints.

PROBLEM 1:

Minimize:

$Z=50X+100Y$

Subject to constraints:

$7x+2y \geq 28$

$2x+12y \geq 24$

x ≥0,y ≥0

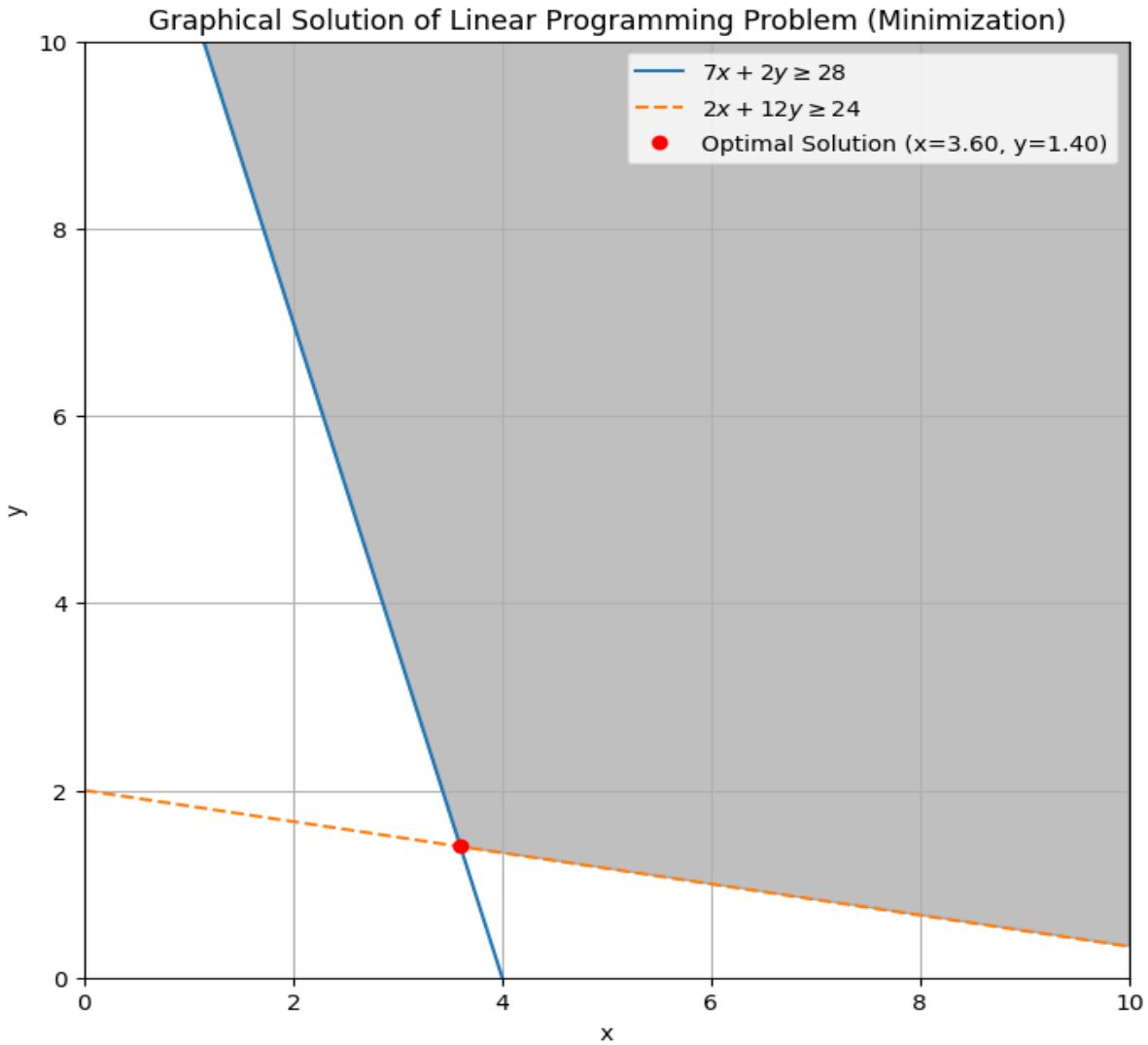
CODE:

```
import numpy as np

import matplotlib.pyplot as plt

from scipy.optimize import linprog

c = [50, 100]

A = [[-7, -2],

[-2, -12]]

b = [-28, -24]

x_bounds = (0, None)

y_bounds = (0, None)

res = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds],
method='highs')

x_opt, y_opt = res.x

print(f"Optimal solution: x = {x_opt}, y = {y_opt}, Z = {z_opt}")

x = np.linspace(0, 10, 400)

y1 = (28 - 7*x) / 2

y2 = (24 - 2*x) /12

plt.figure(figsize=(8, 8))

plt.plot(x, y1, label=r'$7x + 2y \geq 28$')

plt.plot(x, y2, label=r'$2x + 12y \geq 24$', linestyle='--')

y_feasible = np.maximum(y1, y2)

plt.fill_between(x, y_feasible, 10, color='grey', alpha=0.5)

plt.plot(x_opt, y_opt, 'ro', label=f'Optimal Solution (x={x_opt:.2f},
y={y_opt:.2f})')

plt.xlim(0, 10)

plt.ylim(0, 10)

plt.xlabel('x')
```

```
plt.ylabel('y')
plt.title('Graphical Solution of Linear Programming Problem (Minimization)')
plt.legend()
plt.grid(True)
plt.show()
```

EXPECTED OUTPUT:

Optimal solution: x = 3.6, y = 1.4, Z = 320.0



PROBLEM 2:

MAXIMIZE: Z=3x + 2y

Subject to constraints:

x+y≤4

x ≤ 2

y ≤ 3

x ≥ 0, y ≥ 0

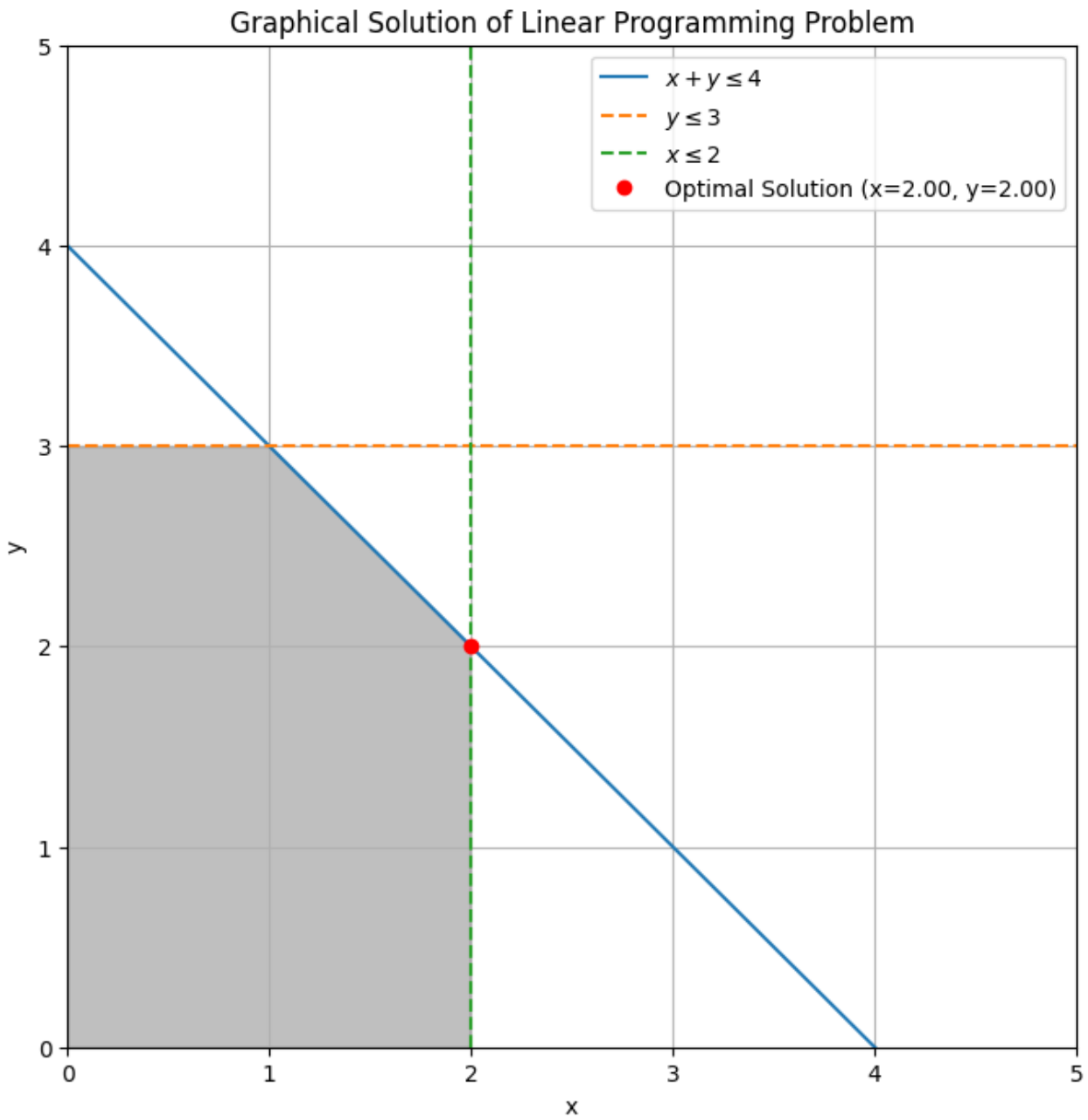
CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import linprog
c = [-3, -2]
A = [[1, 1],
[1, 0],
[0, 1]]
b = [4, 2, 3]
x_bounds = (0, None)
y_bounds = (0, None)
res = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds],
method='highs')
x_opt, y_opt = res.x
print(f"Optimal solution: x = {x_opt}, y = {y_opt}, Z = {z_opt}")
x = np.linspace(0, 5, 400)
y1 = 4 - x
y2 = np.ones_like(x) * 3
y3 = 2 * np.ones_like(x)
plt.figure(figsize=(8, 8))
plt.plot(x, y1, label=r'$x + y \leq 4$')
```

```
plt.plot(x, y2, label=r'$y \leq 3$', linestyle='--')
plt.plot(y3, x, label=r'$x \leq 2$', linestyle='--')
plt.fill_between(x, np.minimum(y1, y2), where=(x <= 2), color='grey',
alpha=0.5)
plt.plot(x_opt, y_opt, 'ro', label=f'Optimal Solution (x={x_opt:.2f},
y={y_opt:.2f})')
plt.xlim(0, 5)
plt.ylim(0, 5)
plt.xlabel('x')
plt.ylabel('y')
plt.title('Graphical Solution of Linear Programming Problem')
plt.legend()
plt.grid(True)
plt.show()
```

EXPECTED OUTPUT:

Optimal solution: x = 2.0, y = 2.0, Z = 320.0



RESULT: Thus, the Linear Programming Problem (LPP) by Graphical method using Python program has been implemented successfully.